

Vancouver's Next Protected Bike Lane

Group: L11

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Outline

- Overview
- Proposed Cycle Lane
- Saliency Model
- Stakeholders
 - Needs and Justification
 - Scoring
- Barrier Decision
- Conclusion
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 - Causal Loop Diagram
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Need for increase in protected bike lanes in Vancouver

Vancouver 2040 Sustainability Goal:

“Make cycling safe, convenient, comfortable and fun for people of all ages and abilities.”

City Trips Made by Cycling

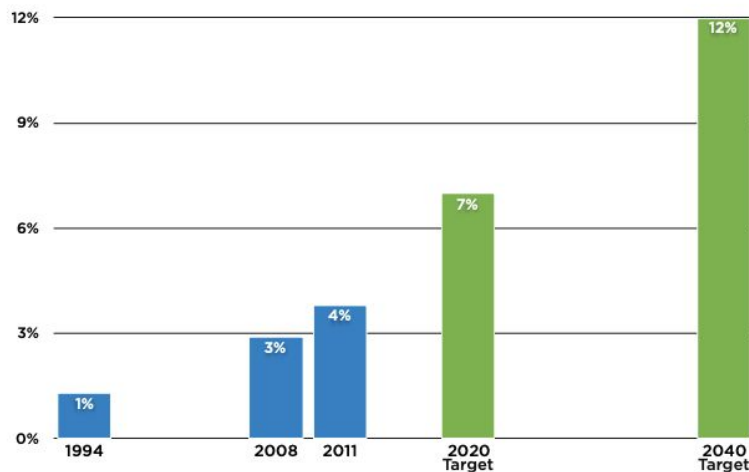


Figure 1: For all trips in the City of Vancouver.
Source: Translink Trip Diaries, City of Vancouver
2020 and 2040 targets.

Potential Cycle Routes



Figure 2: Map of Vancouver
Source: City of Vancouver

Our Proposal: West Kingsway with a Rain Garden



*Figure 3: Segment of West Kingsway, North of Windsor St.
Source: Google Street View.*



*Figure 4: Example of protected bike lane with a rain garden
Source: Meristem Design.*

Saliency Model

Determining Precedence of Stakeholders

- City of Vancouver
- Cyclists
- Businesses
- First Responders
- Car Commuters
- Taxi + Rideshare
- Translink

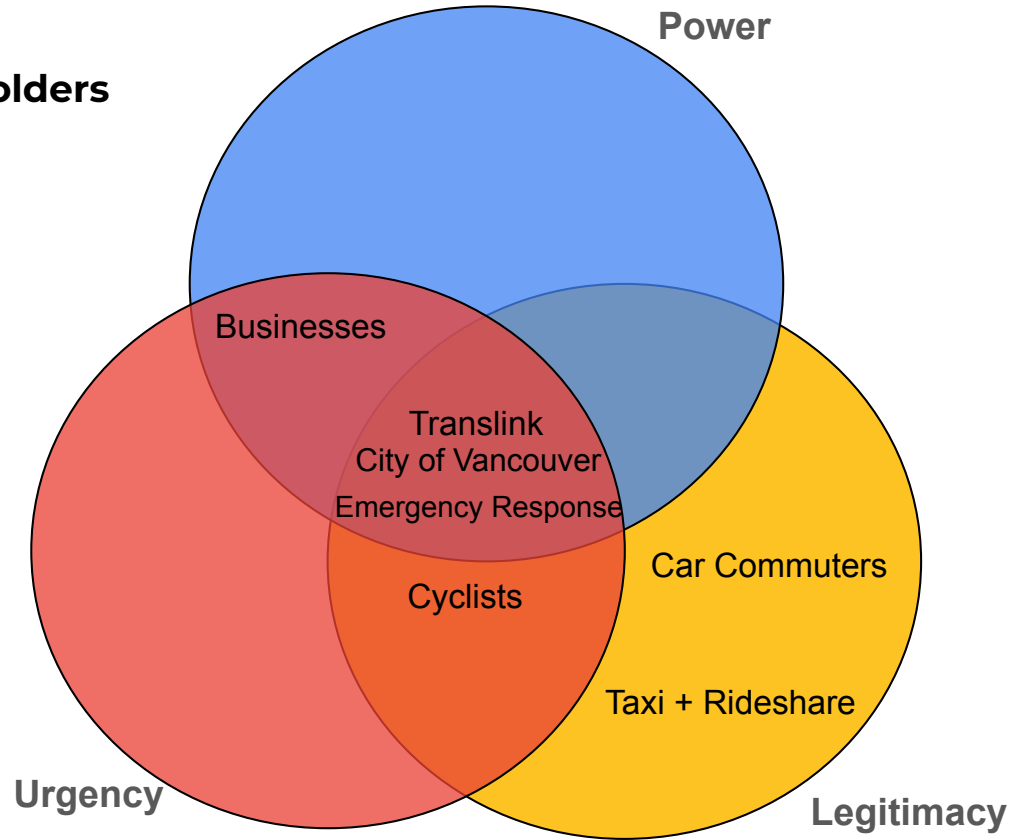


Figure 5: Determining saliency of stakeholders

Emergency Services have Requirements, rather than Evaluation Criteria.

Spatial/Temporal Needs:

- Space for emergency vehicles to access sidewalk.
- Street access for hydrants.
- Timely response to emergencies.

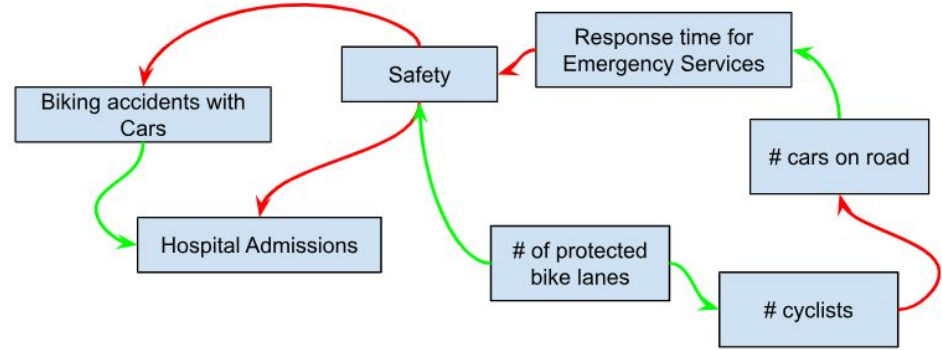


Figure 6. Section of causal loop diagram addressing Emergency Services' needs. (Appendix C)

Drivers' needs are already addressed.

Needs:

- Keep commute times lower
- Minimize air pollution
- Access to street for pickup
- Minimize car-bike accidents
- Parking options

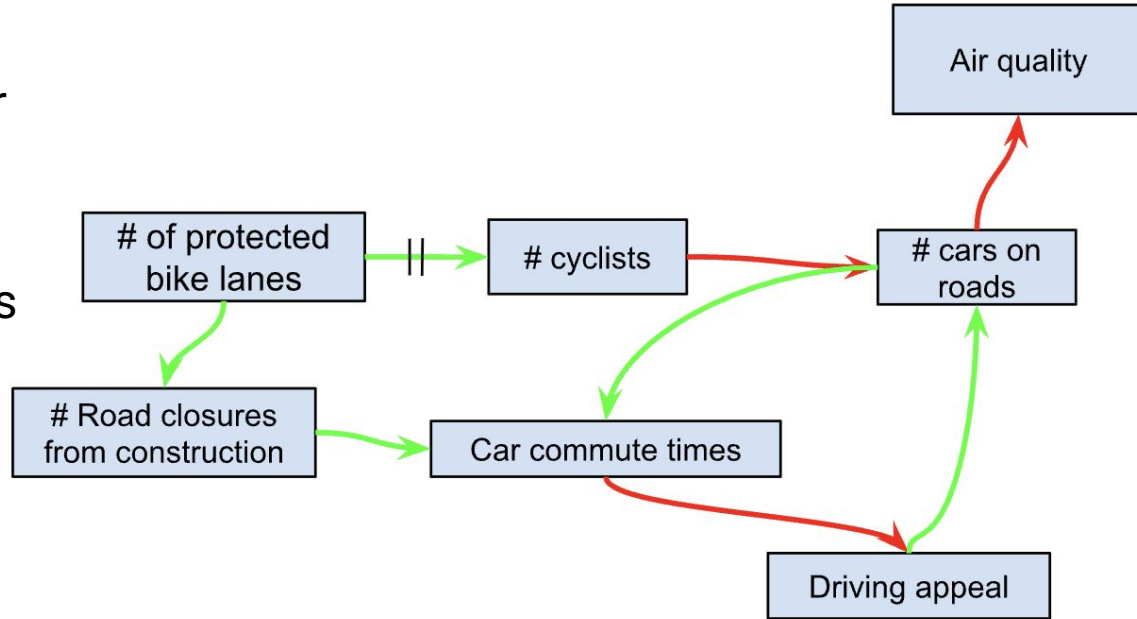


Figure 7. Section of causal loop diagram addressing driver needs (Appendix C)

Allotted Weightings of Stakeholders based on Saliency

Stakeholder Weighting

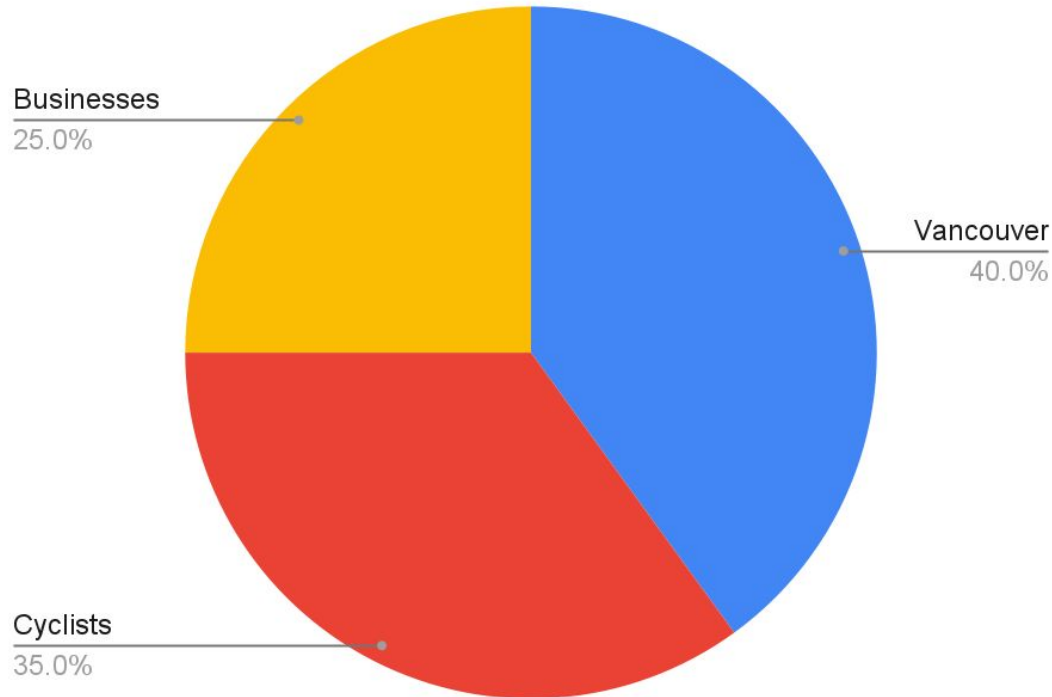


Figure 8: a weighting of each stakeholder

The City of Vancouver's needs are prioritized as they are a primary stakeholder.

- **Economic:**

- Minimize construction time and costs.

- **Social:**

- Safety - Type of barrier.
- Location to benefit most people - based on socioeconomic transportation affordability burden.
- Minimize congestion.

- **Environmental:**

- Air Quality
- Greenhouse gas emission reduction.

The City of Vancouver prefers W. Kingsway location

- Kingsway routes score best in three of four criteria.
- Cost is most important (40%), then usage (30%).
- West 14th Avenue is a very unsatisfactory option.
- Emissions and transport burden align with Transport 2040

WDM Summary for City of Vancouver

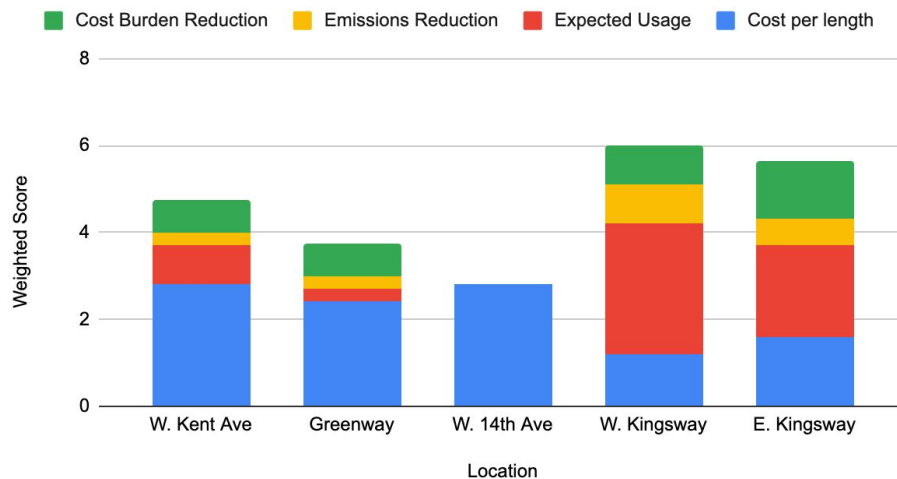


Figure 9: Summarized WDM for City of Vancouver (Appendix A).

Cyclists' needs are addressed as secondary stakeholders.

- Safer bike lanes separated from cars
- Efficient routes with short commute times
- Easy connection to public transport
- Central and accessible corridors
- Bike facilities like racks and storage

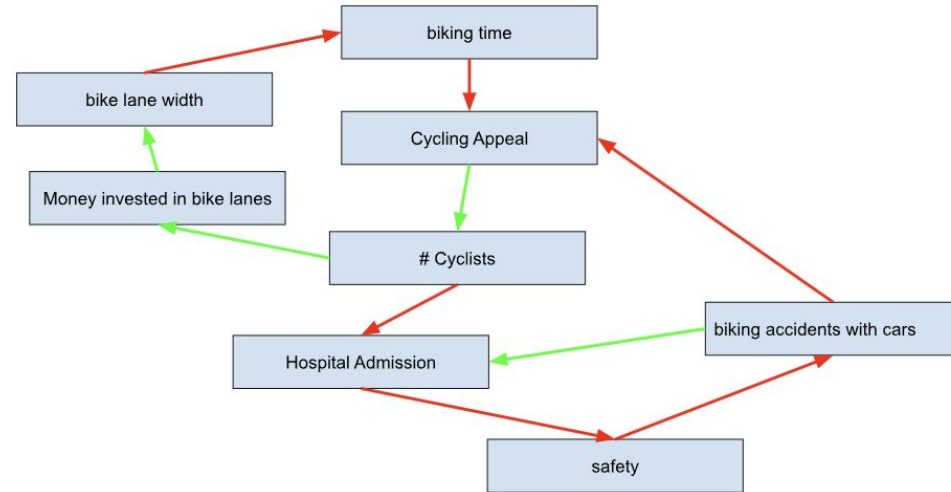


Figure 10. Section of causal loop diagram addressing Cyclists' needs. (Appendix C)

Cyclists prefer West 14th Avenue

- Route safety is most important for cyclists
- Accessible for all Ages Abilities (AAA) designation is important
 - Most cyclist needs are also AAA goals
- Protected lanes improve safety, but roads can still *feel* unsafe
- Rides less than 5km are best for cycling

WDM Summary for Cyclists

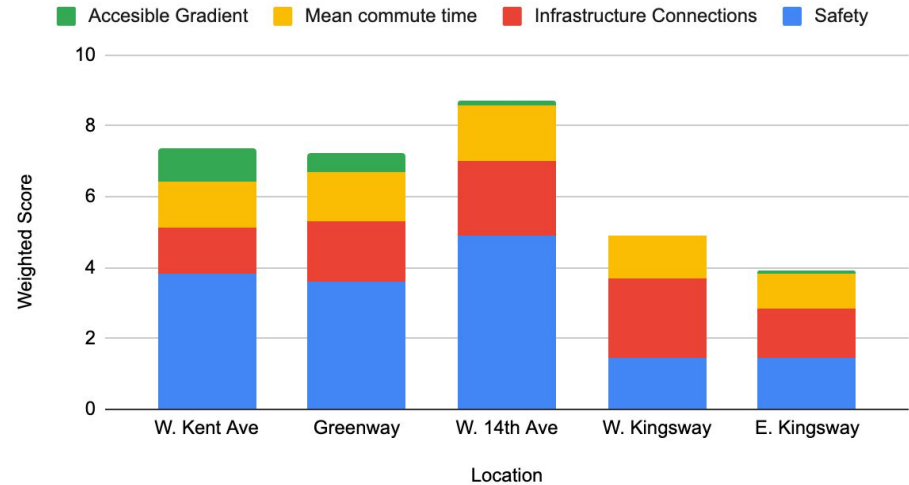


Figure 11: Summarized WDM for Cyclists (Appendix A).

Businesses' needs as secondary stakeholders.

- Parking options for customers is sustained
- Store is accessible for customers
- Bike lane contributes to foot traffic in store

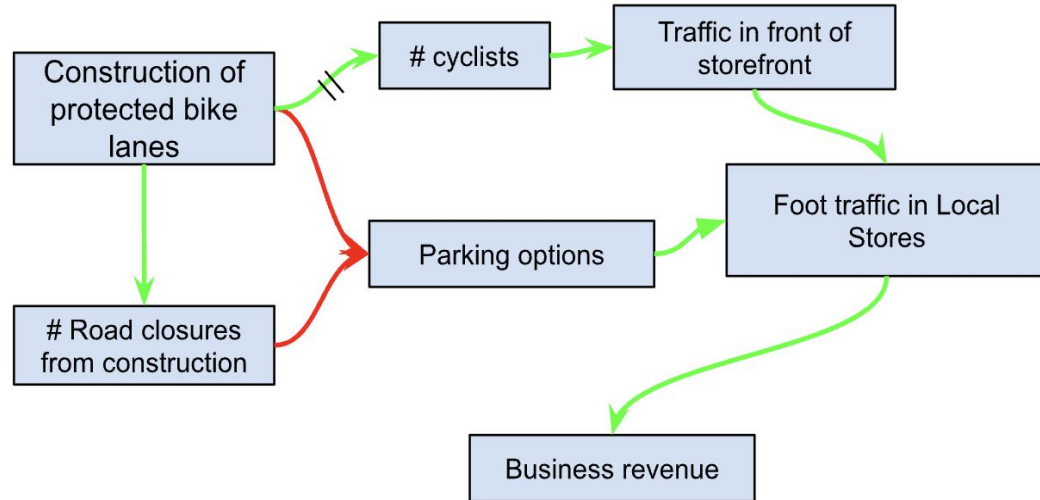


Figure 12. Section of causal loop diagram addressing business needs (Appendix C)

Businesses prefer W. Kingsway location

- Kingsway routes have high business density
- West Kent has unacceptable levels of parking space
- West Kingsway has greater expected cyclist traffic

WDM Summary for Businesses

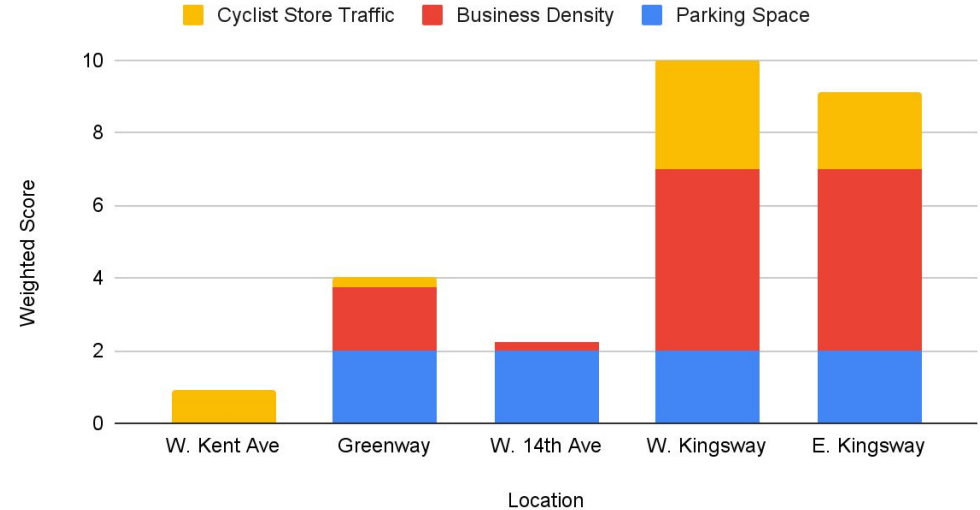


Figure 13: Summarized WDM for City of Vancouver (Appendix A).

Overall Scores based on Saliency of Stakeholders

Overall Results from Saliency Weighting

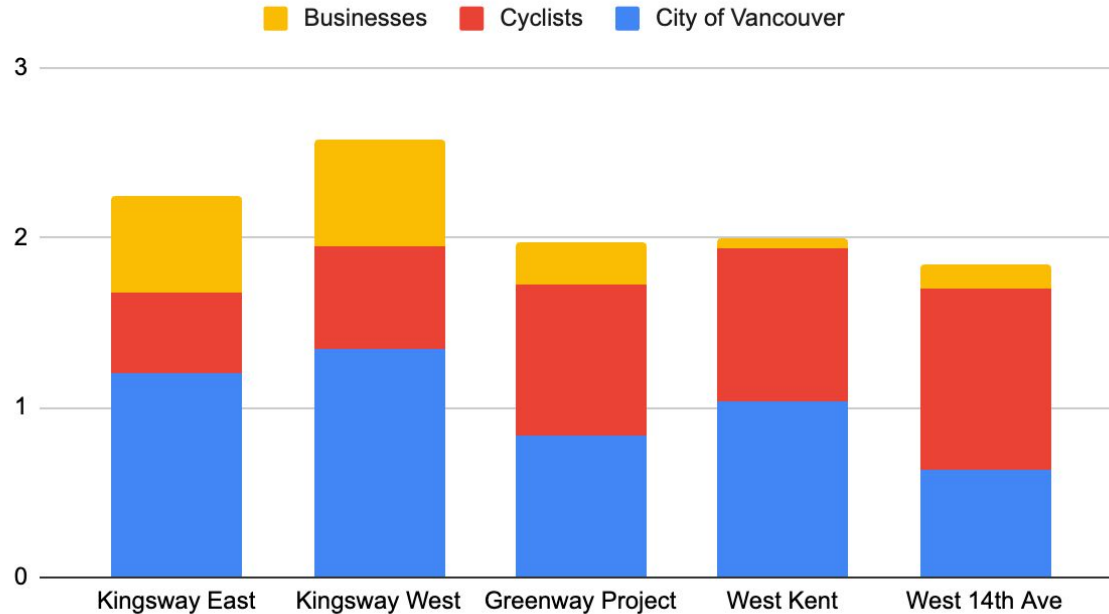


Figure 14: Scored average of each option

Weighted Decision Matrix for Barrier Types

Criterion Weighted Distribution in the WDM

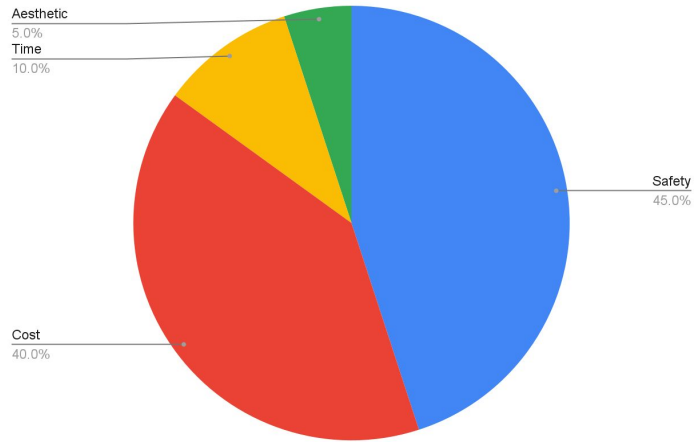


Figure 15: Allotted Weighting for each Barrier Evaluation Criterion

WDM Summary for Barrier Selection

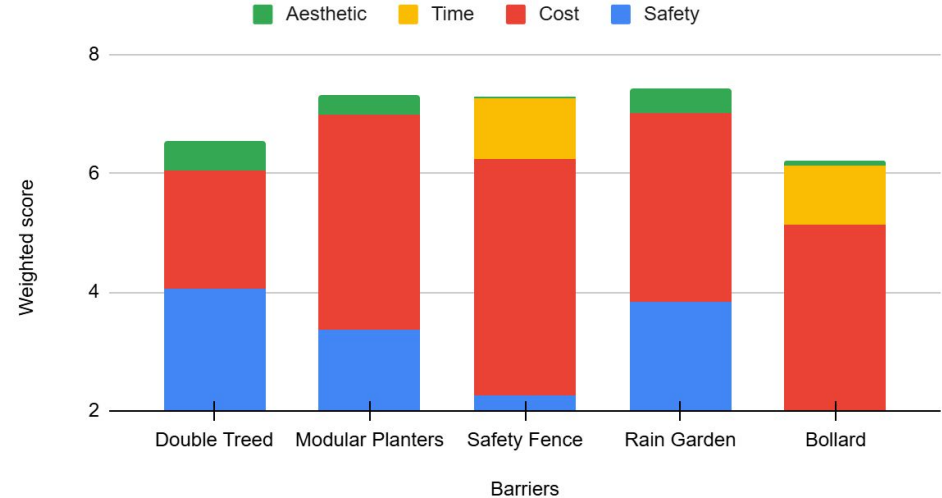


Figure 16: WDM for each Barrier type (Appendix A).

Conclusion: West Kingsway with Rain Garden barrier

Benefits	Limitations	Expected outcomes
Improved connectivity	Short-term traffic disruption	~ 45,000 cyclists per month
Mixed residential and commercial area	Noise pollution	~ \$47 per km per avg-monthly-cyclists
Improved equity	Maintenance requirements	~120 tonnes of CO ₂ reduced per year



Figure 17: Segment of West Kingsway, just north 15th avenue



Figure 18: Example of protected bike lane with a rain garden.

References

City of Vancouver. (2012). Transportation 2040. <https://vancouver.ca/files/cov/transportation-2040-plan.pdf>

City of Vancouver. (2019). Engineering design manual. <https://vancouver.ca/files/cov/engineering-design-manual.pdf>

City of Vancouver. (n.d.). Central Valley Greenway. <https://vancouver.ca/streets-transportation/central-valley.aspx>

City of Vancouver. (n.d.). Walk, bike, roll, transit: Getting around the Vancouver way. <https://vancouver.ca/streets-transportation/walk-bike-and-transit.aspx>

City of Vancouver. (2024). 2024 Vancouver Transportation Survey (VTS) report. <https://vancouver.ca/files/cov/2024-transportation-survey-report.pdf>

City of Vancouver – Open Data Portal. (n.d.). Zoning districts and labels [Data set]. <https://opendata.vancouver.ca/explore/dataset/zoning-districts-and-labels/>

City of Vancouver – Open Data Portal. (n.d.). Intersection traffic movement counts [Data set].
<https://opendata.vancouver.ca/explore/dataset/intersection-traffic-movement-counts/>

Clark, B., & Parkin, J. (2022). Cycling diversion factors: Rapid evidence assessment (Summary report). UK Department for Transport.
<https://assets.publishing.service.gov.uk/media/644a9513c33b46000cf5e380/cycling-version-factors-summary-report.pdf>

Department for Transport. (2023, April 14). Cycling diversion factors. <https://www.gov.uk/government/publications/cycling-diversion-factors>

Google. (n.d.). Google Street View. (Retrieved October 26, 2025.) <https://www.google.com/maps/streetview/>

Meristem Design Ltd. (n.d.). Integrating rain gardens on Forest Road cycle lane [Case study]. <https://www.meristemdesign.co.uk/case-study/forest-road/>

References

Metro Vancouver. (n.d.). Vehicle kilometres travelled (VKT).

<https://metrovancover.org/services/regional-planning/metro-2050-performance-monitoring/goal-5/road-and-vehicle-use/vehicle-kilometres-travelled>

Metro Vancouver. (n.d.). Equity—Regional Growth Management Study, Phase 2: Appendix I (Maps).

<https://metrovancover.org/services/regional-planning/Documents/equity-regional-growth-management-study-phase-2-appendix-i-maps.pdf>

OpenAI. (2025). *ChatGPT* [Large language model]. <https://chatgpt.com/>

Toronto Centre for Active Transportation. (2016). Costing of bicycle infrastructure and programs in Canada.

<https://www.tcat.ca/wp-content/uploads/2016/08/Costing-of-Bicycle-Infrastructure-and-Programs-in-Canada.pdf>

TransLink. (2016). Regional transportation investments: 10-year vision for Metro Vancouver transit and transportation.

<https://www.translink.ca/-/media/translink/documents/plans-and-projects/ten-year-investment-plan/vision/10-year-vision-for-metro-vancouver-transit-and-transportation.pdf>

TransLink. (2022). Rapid implementation design guide for bikeways in Metro Vancouver.

https://www.translink.ca/-/media/translink/documents/cycling/regional-cycling-strategy/rapid_implementation_design_guide_for_bikeways_in_metro_vancouver.pdf

TransLink. (2024). State of cycling in Metro Vancouver 2024.

<https://www.translink.ca/-/media/translink/documents/rider-guide/cycling/reports/translink-state-of-cycling-in-metro-vancouver-2024.pdf>

Vancouver Municipal Code. (2025). § 16.04.160 Amendment to IFC Section 507—Fire protection water supplies. <https://vancouver.municipal.codes/VMC/16.04.160>

The Climbing Cyclist. (2013, May 22). Gradients and cycling: An introduction. <https://theclimbingcyclist.com/gradients-and-cycling-an-introduction/>

Appendix A - WDMs

Barrier Types

		Double Treed Boulevard		Modular Planters		Safety fence		Rain Garden		Bollard	
Criterion	Weight	Raw	Score	Raw	Score	Raw	Score	Raw	Score	Raw	Score
Safety	45%	9	4.05	7.5	3.375	5	2.25	8.5	3.825	2.5	1.125
Cost	40%	5	2	9	3.6	10	4	8	3.2	10	4
Time	10%	0	0	0	0	10	1	0	0	10	1
Aesthetic	5%	10	0.5	7	0.35	1	0.05	8	0.4	2	0.1
	Total		6.55 (#4)		7.325 (#2)		7.3 (#3)		7.425 (#1)		6.225 (#5)

City of Vancouver

		West Kent Ave		Greenway Project		West 14th Avenue		West Kingsway		East Kingsway	
Criterion	Weight	Raw	Score	Raw	Score	Raw	Score	Raw	Score	Raw	Score
Cost per length	40%	7	2.8	6	2.4	7	2.8	3	1.2	4	1.6
Expected Usage	30%	3	0.9	1	0.3	0	0	10	3	7	2.1
Emissions Reduction	15%	2	0.3	2	0.3	0	0	6	0.9	4	0.6
Transport Cost Burden reduction	15%	5	0.75	5	0.75	0	0	6	0.9	9	1.35
	Total		4.75 (#3)		3.75 (#4)		2.8 (#5)		6 (#1)		5.65 (#2)

Cyclists

		West Kent Ave		Greenway Project		West 14th Avenue		West Kingsway		East Kingsway	
Criterion	Weight	Raw	Score	Raw	Score	Raw	Score	Raw	Score	Raw	Score
Traffic Volume	30%	1.5	0.45	9	2.7	10	4.5	1.5	0.45	1.5	0.45
Lighting	10%	10	1	9	0.9	4	0.4	10	1	10	1
Bike Lane Connectivity	15%	4	0.6	4	0.6	10	1.5	8	1.2	4	0.6
Commute time of Residence in Area	20%	5	1	7	1.4	8	1.6	6	1.2	5	1
Accessible with Mobi Bike	5%	0	0	10	0.5	6	0.3	5	0.25	0	0
Slope Gradient	10%	0.6	0.06	5.4	0.54	1.3	0.13	0	0	0.6	0.06
Integration with Transit	10%	8	8	6	0.6	3	0.3	8	0.8	8	0.8
	Total		3.91		7.24		8.73		4.9		3.91

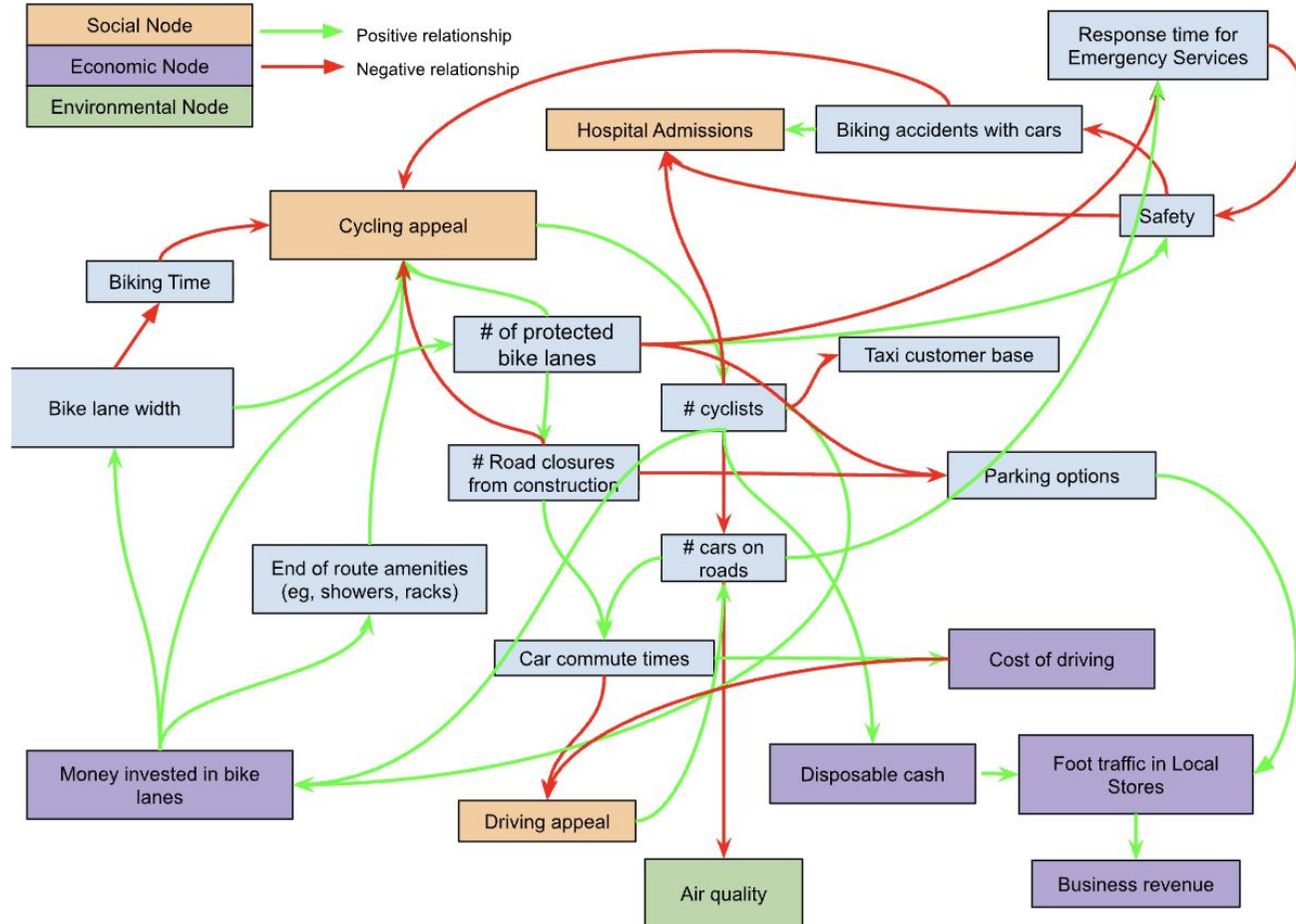
Businesses

		West Kent Ave		Greenway Project		West 14th Avenue		West Kingsway		East Kingsway	
Criterion	Weight	Raw	Score	Raw	Score	Raw	Score	Raw	Score	Raw	Score
Parking Spaces	20%	0	0	10	2	10	2	10	2	10	2
Economic Business Density	50%	1.75	0	3.5	1.75	0.5	0.25	10	5	10	5
Expected Number of Monthly Cyclists	30%	0.3	3	1	0.3	0	0	10	3	7	2.1
	Total		0.9		4.05		2.25		10		9.1

Appendix B - Overall Stakeholder Weighting

		West Kent Ave		Greenway Project		West 14th Avenue		West Kingsway		East Kingsway	
Criterion	Weight	Raw	Score	Raw	Score	Raw	Score	Raw	Score	Raw	Score
City of Vancouver	40%	5.16	2.07	4.16	1.67	3.18	1.27	6.75	2.7	5.99	2.40
Cyclists	35%	7.38	2.58	7.24	2.53	8.73	3.06	4.9	1.72	3.91	1.37
Businesses	25%	0.9	0.23	4.05	1.01	2.25	0.56	10	2.5	9.10	2.28
	Total		4.87		5.21		4.89		6.91		6.04

Appendix C - Causal Loop Diagram



Usage of Generative AI

OpenAI's ChatGPT-5 was used to draft the formatted references list, after prompting it with the links of the sources that were used.

Otherwise, no generative AI tools were used to generate the content or images of any slide, the scoring guidelines, data for use in WDMs, or in any other way throughout this module.